

Module 12

Configuring Wireless Networks

Learning Objectives

- Summarize wireless standards.
- Install and configure secure wireless networks.
- Troubleshoot wireless networks.

| Lesson 12.1

Wireless Concepts and Standards

802.11 Wireless Standards

- Commonly known as Wi-Fi
- Established by the Electrical and Electronics Engineers (IEEE)
- Defines how radio waves communicate over distances
- Ensures compatibility between wireless devices



802.11 Characteristics

Physical Layer

- Encodes data into radio signals with various modulation schemes

Media Access Control (MAC)

- Uses CSMA/CA for efficient transmission and to minimize collisions

Topologies

- Primarily a logical star
- Centered around an Access Point (AP) connecting wireless devices

Evolution

- 1 Mbps in the initial standard
- Over 1 Gbps in modern standards

Wi-Fi Alliance

- Certifies products for standard adherence and device compatibility



802.11a

Frequency Band

- Operates in the 5 GHz band
- Avoids the crowded 2.4 GHz band used by many household devices

Data Rate

- Nominal data rate of up to 54 Mbps

Technology

- Uses Orthogonal Frequency-Division Multiplexing (OFDM) for efficient data transmission

Pros and Cons

- Less prone to interference
- Shorter range compared to technologies operating in the 2.4 GHz band

802.11b/g

802.11b

- **Frequency Band**
 - Remains in the 2.4 GHz band
- **Data Rate**
 - Increases to a nominal 11 Mbps
- **Technology**
 - Direct Sequence Spread Spectrum (DSSS) and Complementary Code Keying (CCK) for signal encoding
- **Channel Overlap**
 - Potential for co-channel interference due to overlapping channels

802.11g

- **Compatibility**
 - Backward support for 802.11b while offering increased data rates
- **Frequency Band**
 - Uses the 2.4 GHz band with the same channel layout as 802.11b
- **Data Rate**
 - Offers a nominal data rate of 54 Mbps using OFDM

802.11n

Frequency Bands

- Supports both 2.4 GHz and 5 GHz bands
- Accommodates wider channel bandwidth
- Reduced interference

Technology Enhancements

- Multiple Input Multiple Output (MIMO) and Channel Bonding significantly increases bandwidth and reliability
- Multiple antennae send and receive up to four separate data streams, enhancing signal reliability and range
- Combines two adjacent 20 MHz channels into a single 40 MHz channel for increased data rates

Data Rate

- Achieves up to 72 Mbps per stream
- Potential rates up to 600 Mbps under optimum conditions

Wi-Fi 5 (802.11ac)

Frequency Band

- Operates exclusively on the 5 GHz band

Throughput

- Up to Gigabit speeds with 80/160 MHz channel bonding

Spatial Streams

- Supports up to 8 spatial streams for enhanced data rates

Modulation

- Uses denser modulation at close ranges for higher data throughput

Marketing Labels

- Devices marketed using AC values like AC5300 (combined throughput capacities)

Multiuuser MIMO

MU-MIMO Introduction

- Enhances network efficiency
- Allows simultaneous multiple user access

DL MU-MIMO

- Multiple antennae AP sends data to multiple stations at once

Wi-Fi Generations

- Wi-Fi 5 and 6 support parallel station communications, extending up to 8 stations

Benefits

- Better bandwidth and network efficiency
- Improved performance
- Supports more devices in crowded areas

Band Steering

Purpose

- Optimizes dual-band/tri-band Wi-Fi usage
- Prioritizes connection to 5/6 GHz bands over 2.4 GHz

Function

- Uses beacon control to manage band selection
- Reduces 2.4 GHz band signals to steer devices towards 5/6 GHz bands

Benefits

- Improve network efficiency and performance
- Can be reserved for devices without 5/6 GHz capacity

Cellular Technologies

Feature	2G/3G	4G (LTE/LTE-A)	5G
Peak Data Rate	2G: Up to 384 Kbps 3G: Up to 2 Mbps	LTE: Up to 150 Mbps LTE-A: Up to 300 Mbps	Up to 20 Gbps
Technology Base	2G: Digital voice and SMS 3G: Mobile internet	High-speed mobile Internet	Ultra-high-speed internet, IoT, massive MIMO
Spectrum Use	Limited band use	Efficient use of spectrum	Uses a broad spectrum, low to high bands (sub-1 GHz to 40 GHz)
Typical Use Case	Voice calls, basic internet	Streaming, gaming, HD video	Streaming in 4K/8K, AR/VR, smart cities, autonomous vehicles

Satellite Internet Technologies

Microwave Satellite Broadband

- Direct or relay via microwave dishes to orbiting satellites
- Popular for home & expanding in business, especially in remote areas

Geostationary Orbital Satellites

- Positioned in high orbit
- Provide broad coverage
- High latency limits real-time applications

Low Earth Orbital (LEO) Satellites

- High bandwidth, low latency (requires dynamic alignment)
- Closer orbit offers better performance, needs dynamic tracking

Other Satellite Applications

Global Positioning System (GPS)

- Satellite positioning, enhanced by A-GPS for faster fixes
- GPS positioning with A-GPS cellular data boost

Satellite TV and Radio

- Broadcast directly to consumers
- Provide a wide range of entertainment and information channels

Earth Observation and Monitoring

- Satellites equipped with sensors
- Support environmental monitoring, weather forecasting, and disaster management

Activity: Fill in the Blank

1. The _____ GHz band is used by many household appliances.
2. The 802.11 _____ incorporates MIMO and channel bonding to increase bandwidth and reliability
3. _____ prioritizes connection to 5/6 GHz bands over 2.4 GHz bands.



Activity: Fill in the Blank

1. The 2.4 GHz band is used by many household appliances.
2. The 802.11 N incorporates MIMO and channel bonding to increase bandwidth and reliability
3. Band Steering prioritizes connection to 5/6 GHz bands over 2.4 GHz bands.



Lesson 12.2

Enterprise Wireless Network Design

Wireless Local Area Network (WLAN)

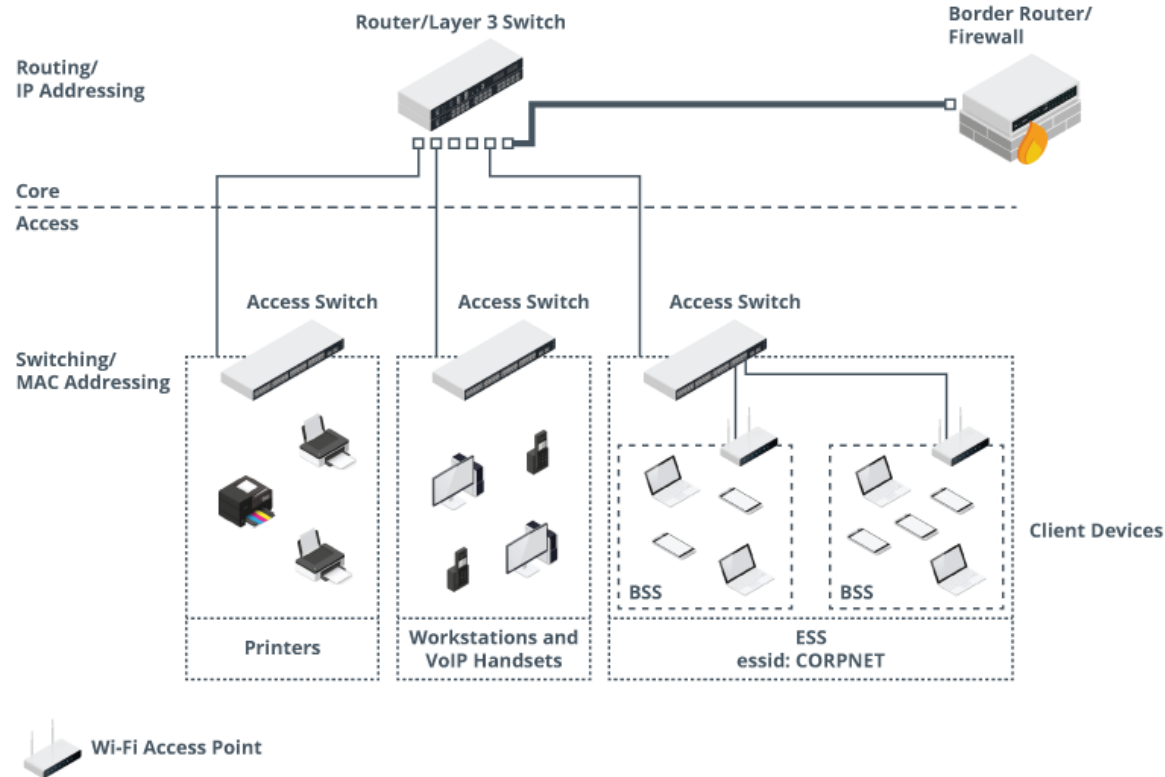
Wireless Local Area Network (WLAN)

- Devices communicate over a wireless signal
- Uses a network name or Service Set Identifier (SSID) for connection

Infrastructure Mode

- Access Point (AP) mediates communications
- Forms a logical star topology
- AP supports multiple BSSs on different bands with unique/shared SSIDs

WLAN Example



BSSID and ESSID

Basic Service Set Identifier (BSSID)

A unique identifier for access points within a WLAN

The "address" for efficient network management

Extended Service Set Identifier (ESSID)

Network name for multiple BSSs within larger WLAN network

Supports client roaming across channels/bands

Range and Signal Strength

Wi-Fi Range

Indoor: ~30m/100ft

Outdoor: 2-3x
indoor range

2.4 GHz > 5 GHz for
range

802.11n+ for better
range

Dynamic Rate Switching

Adjusts data rates
by signal quality

High rates for
strong signals

Interference and Obstructions

Solid
objects/electronics
reduce signal

Concrete/metal
very challenging

Signal Strength Measurement

Measured in dBm;
closer to 0 dBm =
better

Ideal: ~-30 dBm;
Good: ~-65 dBm;
<-80 dBm = packet
loss

RSSI and SNR

RSSI: Closer to 0
dBm = better

SNR: Higher values
= clearer signals

Wireless Survey

Definition

- Also known as a WLAN or RF site survey
- Wireless network planning for optimal coverage, bandwidth, and quality of service

Purpose

- Determine the optimal placement of wireless access points
- Identify potential interference sources

Process

- Physical site inspection
- Signal strength measurement
- Data analysis for network planning

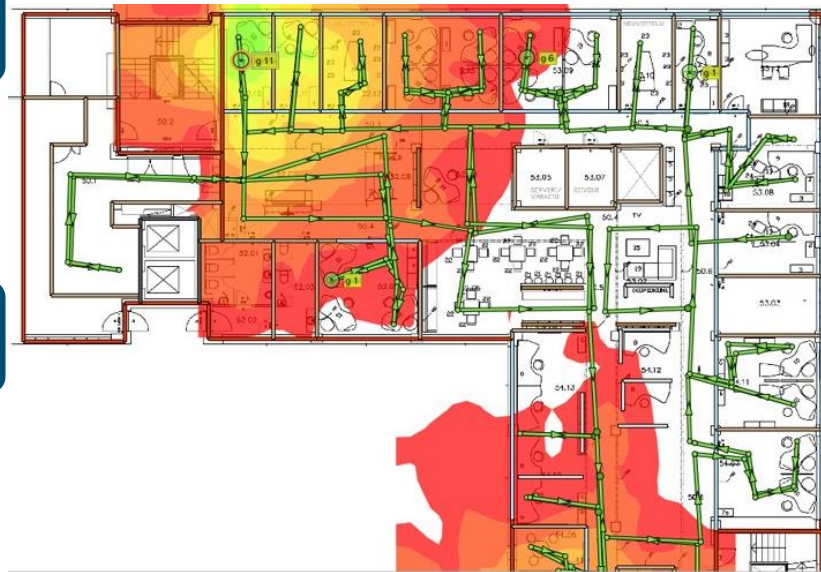
Heat Maps

Definition

- A visual representation of data
- Values depict color showing the intensity of signal strength in an area

Application

- Identify hotspots of network activity
- Identify areas where signal enhancement is needed



Wireless Roaming

Definition

- Seamless device connection across APs without disconnects

Extended Service Area (ESA)

- Network of APs with the same ESSID and security settings

Seamless Transition

- Devices reassociate with new APs based on signal strength, potentially needing reauthentication

Challenges

- Needs balance in signal quality assessment and reassociation rate

SSID Broadcast and Beacon Frame Essentials

SSID Broadcast

- Facilitates connection to WLAN by advertising its presence
- Configurable to enhance security by suppressing broadcast

Beacon Frames

- Special frames that carry essential network information like SSID/ESSID, BSSID, and security protocols
- Broadcast interval adjustable for network performance

Wireless Distribution System (WDS)

Purpose

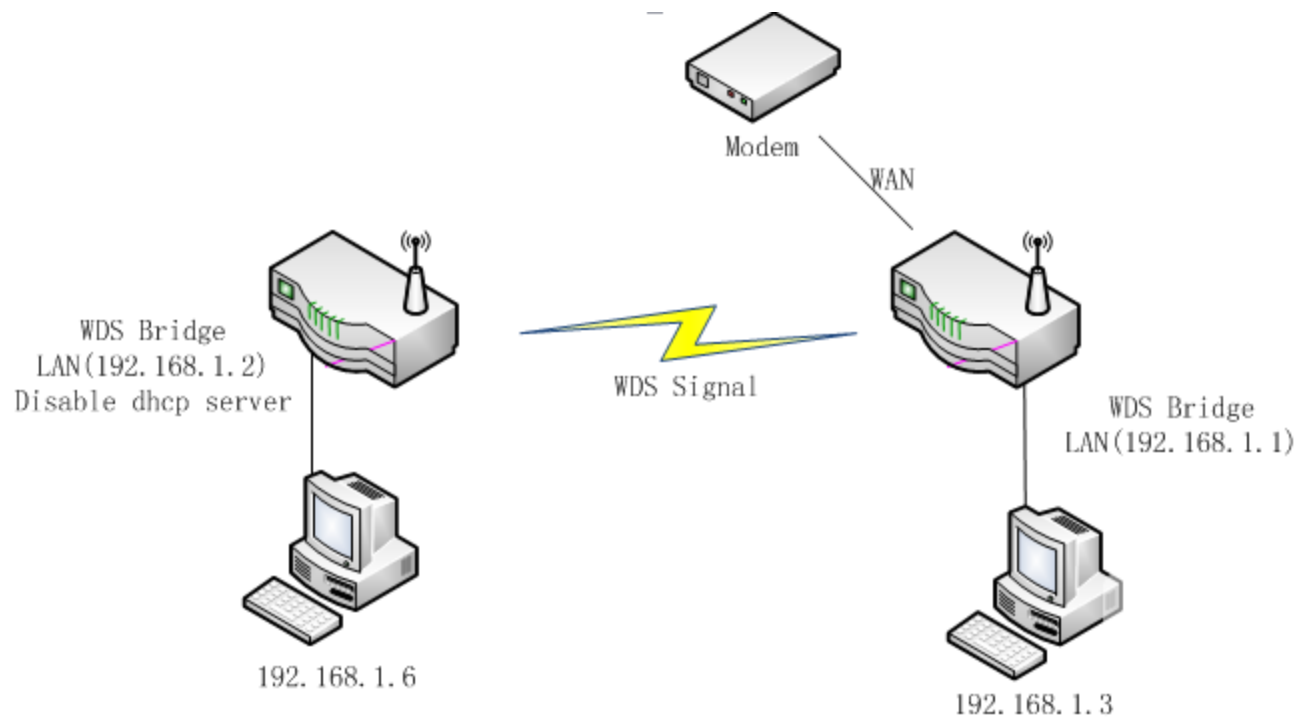
- Creates a wireless network where cabling is impractical

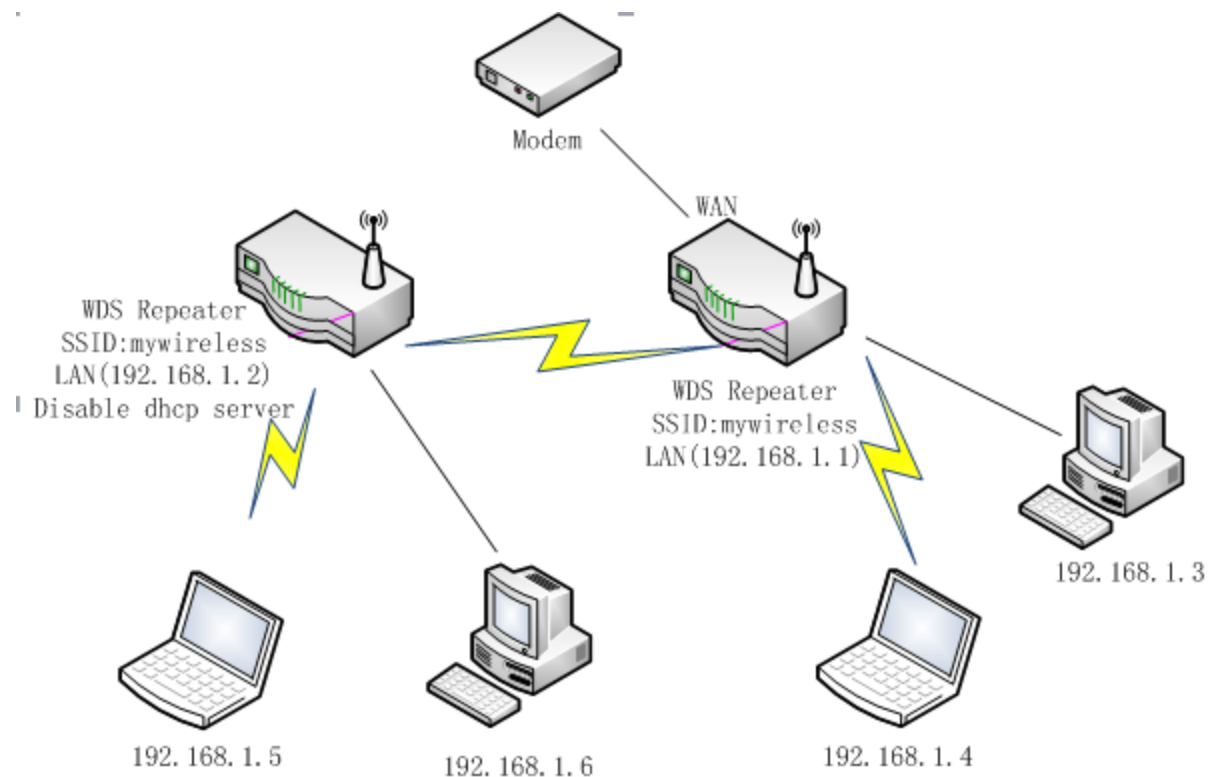
Configuration Requirements

- Matching SSID, channel, and security settings across APs
- Base station (wired) and remote stations (wireless extension)

Vendor Considerations

- Same-manufacturer APs may offer better compatibility and performance





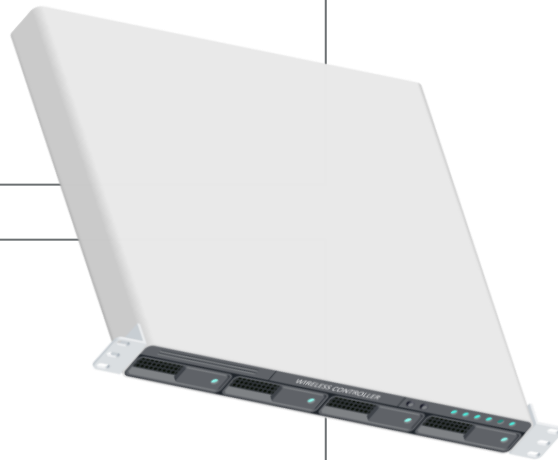
Wireless Controllers

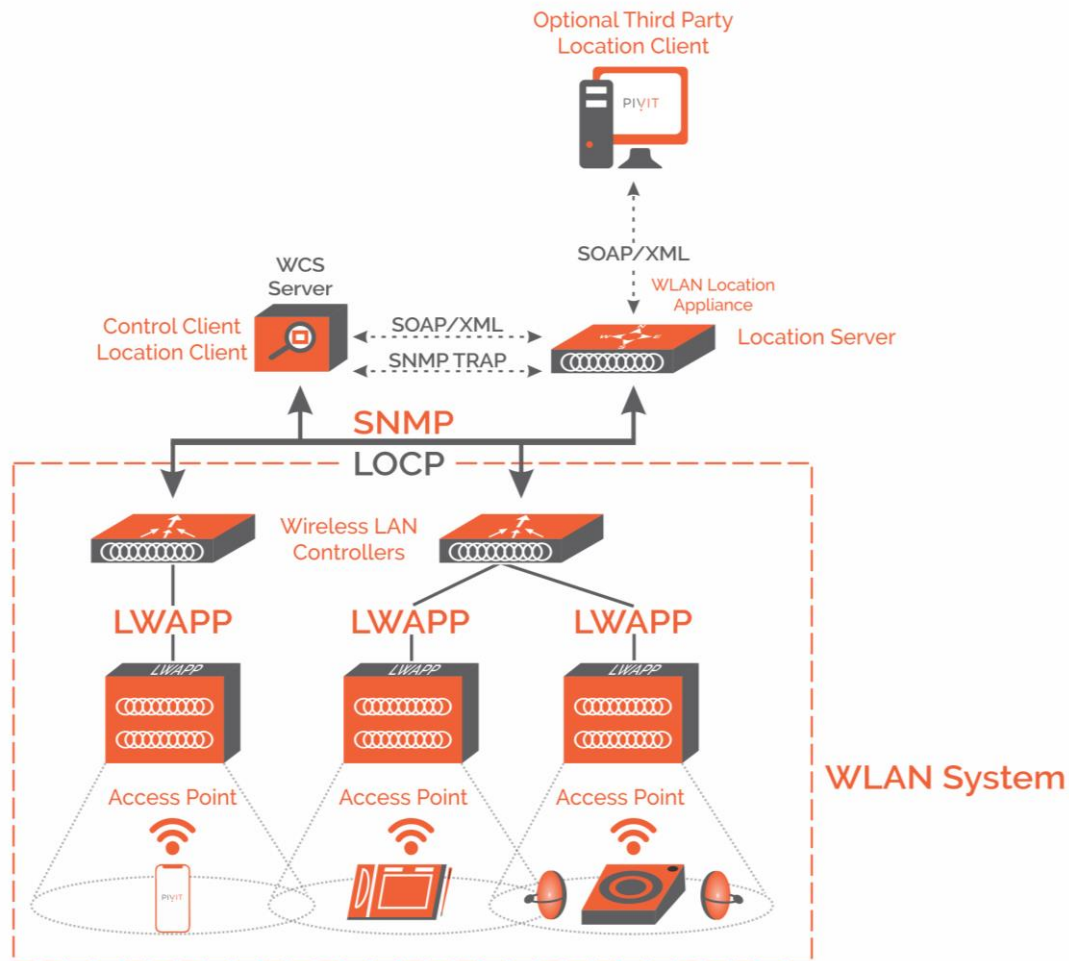
Wireless Controllers

- Manage and monitor multiple APs
- Simplify network configuration
- Prevent individual AP errors
- Offer a comprehensive view of network performance

Functions

- Central management
- Supports up to 1,500 APs and 20,000 clients at once
- Serves as a central point for switching and routing
- Automatically configures SSID, channel, and security settings
- Assigns clients to separate VLANs
- Regulates station numbers per VLAN to minimize broadcast traffic





Antennas

Omnidirectional Antennas

- Radiates and receives signals in all directions equally
- Wide area coverage: Ideal for spreading the signal across large spaces (ceiling-mounted)

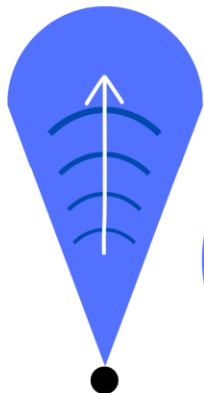
Unidirectional Antennas

- Focuses signal on a specific direction
- Point-to-point communications, extending signals to remote areas
- Provides increased signal strength in the focused area, requires precise alignment

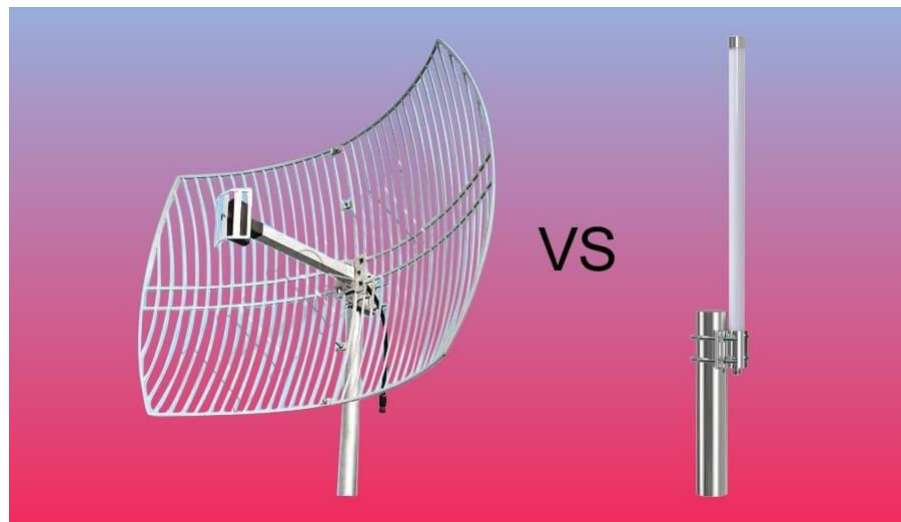
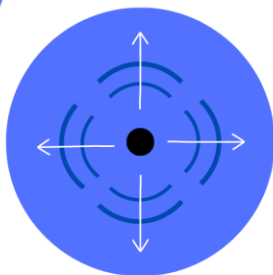
Dual-Polarized Antennas

- Transmits and receives signals in multiple orientations
- Supports mobile devices well, adaptable to various device orientations
- Ensures robust signal quality for devices in different positions

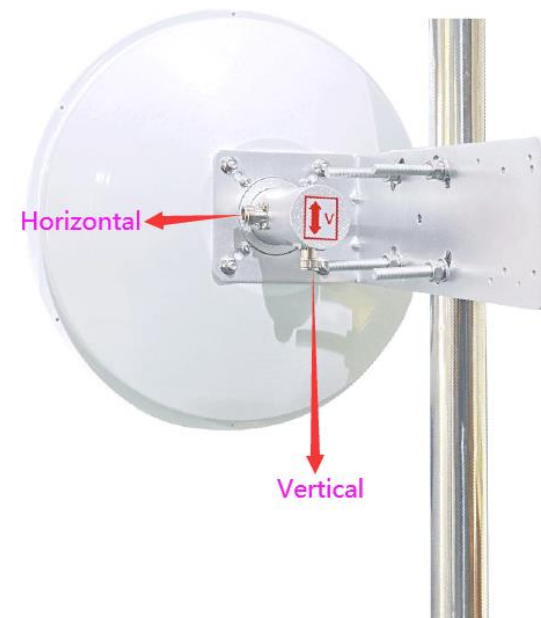
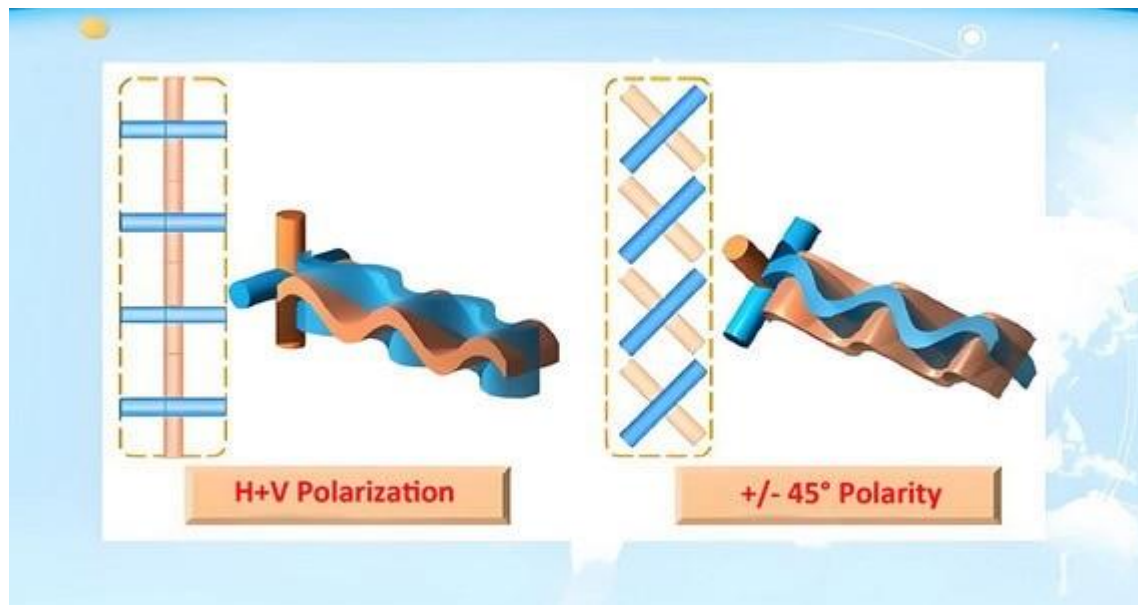
Directoinal Antenna



Omnidirectoinal Antenna



Dual-Polarized Antennas



Activity: What is it?

Antenna that radiates and receives signals in all directions equally

General coverage, suitable for environments where the signal needs to be spread across a wide area

Offers wide coverage, best mounted on ceilings to optimize reach

Answer:

- A. Unidirectional
- B. Omnidirectional
- C. Dual-Polarized

Activity: What is it?

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Answer:

- A. Unidirectional
- B. Omnidirectional**
- C. Dual-Polarized

| Lesson 12.3

Wireless Security

Wi-Fi Protected Access 2 (WPA2)

Background

- WPA2 is the second generation (2004) of Wi-Fi security protocols
- Significantly improved upon WEP and WPA protocols

Encryption Strength

- Relies on shared encryption key for all network connected devices
- Provides robust protection against common Wi-Fi attacks

Security Protocols

- Uses the Advanced Encryption Standard (AES) for data protection
- Remains susceptible to interception and other attacks

Wi-Fi Protected Access 3 (WPA3)

Background

- Introduced in 2018 to address the limitations of WPA2

Individualized Data Encryption

- Implements individualized data encryption for each device

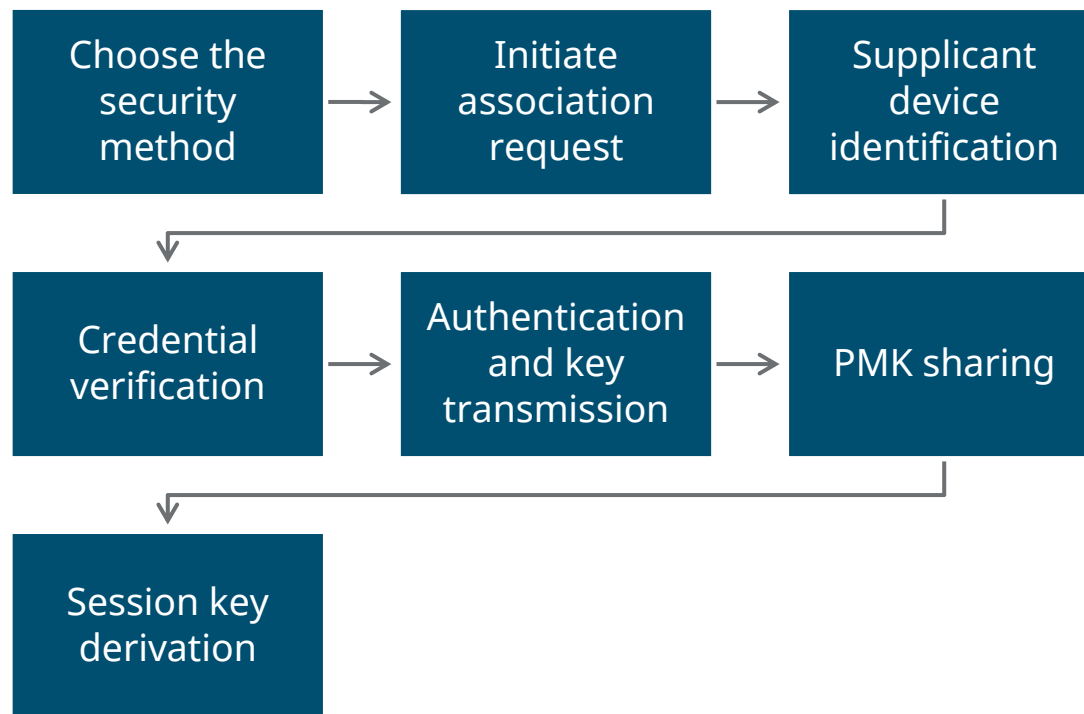
Password Protection

- WPA3 makes password cracking much harder
- Attackers must interact with your Wi-Fi for every password guess, making brute force attacks almost impossible

Forward Secrecy

- Supports forward secrecy
- If an attacker learns your password later, they can't decrypt previously captured data—only newly captured data

Enterprise Authentication



Guest Networks

Defined

- Offer separate access for visitors and traffic isolated from the main network

Benefits

- Enhanced Security: Segregating guest user access from internal resources

Accessibility

- Simplify connection for guests while maintaining network integrity

Implementation

- Separate SSIDs: Distinct from the main network for easy identification
- Restricted Access: Limits to Internet only (no entry to local servers/sensitive data)

Captive Portals

Act as a secondary authentication step in open networks by requiring users to authenticate through a web page

Features

HTTPS Protocol	Secures login information and meets modern browser security standards
Digital Certificate	Ensures the portal is trusted by the client's browser, mitigating redirect issues

Best Practices

Clear Terms and Conditions	Inform users about network use policies and any charges
VPN Recommendation	Encourage the use of VPNs for secure browsing over open or guest networks

Issues with BYOD

Compatibility & Support

- Complex connectivity across devices/OS
- Ensuring seamless network functionality

Security Concerns

- Varied device security levels
- Unpatched devices as threats
- Limited IT control
- Insider threats
- Data vulnerability

Risk Mitigation

- Implement EMM suites
- Use corporate workspaces
- Enforce security policies
- Segregate corporate/personal data

Rogue Access Points

Definition

- Evil twin: fraudulent Wi-Fi AP mimicking a legitimate network
- Users unknowingly connect, allowing attackers to intercept data

How They Work

- Rogue APs deceive users by appearing genuine
- Attackers capture login details, track connections, install malware

Mitigation

- Regularly scan for rogue access points
- Educate users about risks; avoid connecting to suspicious networks

Deauthentication Attacks

Definition

- Disrupt connections between users and Wi-Fi access points
- Attackers force devices to lose access and then reconnect to a network they control

Purpose

- Disrupt communication
- Attackers track connections, capture data, trick users into installing rogue programs

Detection and Prevention

- Monitor network traffic for unusual deauthentication patterns
- Implement strong encryption and authentication mechanisms

Evil Twin Network

Definition

- Fraudulent Wi-Fi APs lure users into connecting to them instead of legitimate ones
- Steal personal data, insert malware, or compromise devices

Function

- Attackers set up fake Wi-Fi access points
- Users unknowingly connect, thinking it's a legitimate network

Identification

- Be cautious in public places with open Wi-Fi networks
- Verify network names and use VPNs for added security

Activity: Two Truths and a Lie

WPA2 uses
the Advanced
Encryption Standard
(AES) for data
protection

WPA2
supports forward
secrecy

WPA2 relies on
a shared encryption
key for all devices
connected to the
network



Activity: Two Truths and a Lie

WPA2 uses
the Advanced
Encryption Standard
(AES) for data
protection

WPA2
supports forward
secrecy

LIE

WPA2 relies on
a shared encryption
key for all devices
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| Lesson 12.4

Wireless Troubleshooting

Wireless Performance Assessment

Important Metrics

Bit Rate

Throughput

Understanding RF Attenuation

Signal degradation with distance governed by the inverse-square rule

Impact of distance and interference on signal strength measured in dB

Concepts in Wireless Connectivity

Signal-to-Noise Ratio (SNR)

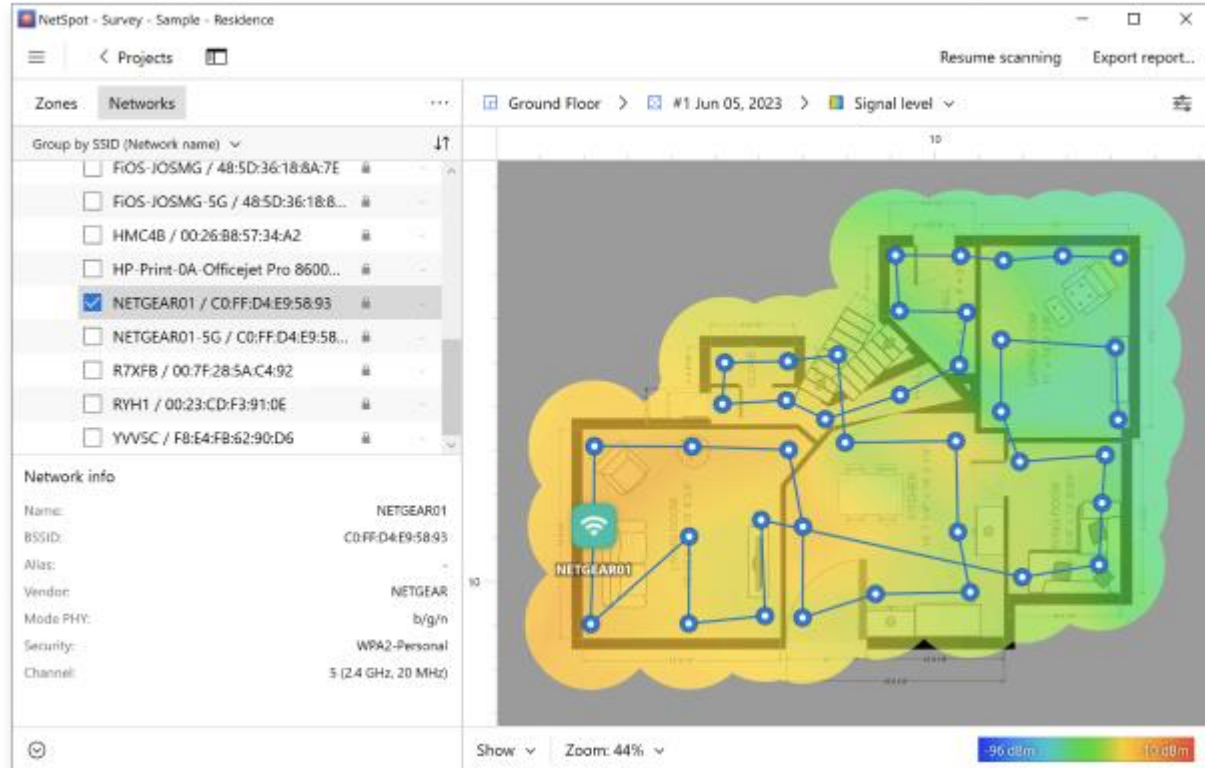
Optimal SNR Margins

Tools for Assessment

Wi-Fi Analyzers: real-time signal and noise measurements

Dedicated Wi-Fi Tester Hardware: comprehensive network analysis

WIFI Analyzer



Insufficient Wireless Coverage Issues

Definition

- Areas with no/poor Wi-Fi signal strength
- Results in connectivity issues within a facility

Causes

- Signal blockage due to physical obstacles
- Greater than optimal distance from access points
- Electronic interference from nearby devices

Insufficient Wireless Coverage Solutions

Extending Coverage	Antenna Optimization	Power Adjustment	Considerations	Goal
• Install more access points • Use range extenders or wireless bridges	• Site surveys for ideal placement • Appropriate antenna types	• Calibrate AP transmit power to match weakest client device	• Compatibility with client device wireless standards • Operational frequency bands (optimal performance)	• Enhance signal reach/quality • Ensure effective wireless communication

Channel Overlap Issues

Channel Overlap

- Occurs with close access points using similar frequencies
- Causes interference and reduces network performance

Types of Overlap

- Co-channel Interference (CCI): Devices compete for the same channel
- Adjacent Channel Interference (ACI): Devices on overlapping channels slow communication

Channel Overlap Solutions

Design Strategies

- Limit channel use to 50% or less for no ACI
- Maintain 25 MHz spacing between channels for no ACI
- Optimal 2.4 GHz channels: 1, 6, 11

Power Adjustments

- Lower AP power to reduce interference
- Avoid maximum power to prevent one-way communication problems

Interference Issues

CCI & ACI

- Multiple devices on same or close frequencies

Physical Blocks

- Walls, furniture, people
- Bounce signals, lowering strength and causing issues

EMI

- Devices in same frequency (microwaves, Bluetooth)
- Disrupts signals

Interference Solutions

Solutions

- Place AP strategically to dodge physical blocks and boost signal
- Change frequency or channel to cut CCI/ACI

Spectrum Tools

- Use to find and fix electromagnetic interference for better signal

Overcapacity Issues

Definition of Overcapacity

- Too many devices attempt to connect to single AP
- Leads to network congestion

Impact on Performance

- Bandwidth saturation
- Slow speeds or unreliable connections for users

Common Causes

- High density of client devices in a limited area
- Single AP bearing too many connections beyond capacity

Effects of Device Saturation

- Includes degraded service, slower web browsing
- Potential bottlenecks moving upstream to the WAN

Overcapacity Solutions

Strategic AP Placement

- Distribute multiple APs to spread out client device connections evenly.

Client Limit Configuration

- Set a maximum number of devices that can connect to each AP to prevent overloading

Use of Traffic Shaping Tools

- Implement traffic shapers to manage bandwidth allocation and prioritize essential services

Dynamic Channel Assignment

- Use technology to dynamically adjust channels and reduce interference and overuse

Continuous Monitoring and Management

- Employ enterprise Wi-Fi solutions with advanced diagnostics to identify and mitigate overcapacity issues proactively

Summary

- Define **network requirements**
- **Survey site** with floor plan & Wi-Fi analyzer
- **Determine AP range** for chosen technology
- **Test installation:** size, security, functionality (real-world conditions)